MIL-P-52192B(MR)
5 January 1971
SUPERSEDING
MIL-P-52192A(MR)
16 March 1967

MILITARY SPECIFICATION

PRIMER COATING, EPOXY

1. SCOPE

1.1 Scope. This specification covers an air-drying, or baking chemical-resistant, epoxy-resin primer. It provides for two compositions, one of which is suitable for use under AIR POLLUTION REGULAT'ONS (see 6.5).

1.2 Classification. Primer covered by this specification shall be of the following composition as specified.

Composition G - General use.
Composition L - Limited use (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-B-846 - Butyl Alcohol, Normal (Butanol) (for Use in Organic Coatings).
TT-E-516 - Enamel, Lustreless, Quick-Drying Styrenated Alkyd Type.
TT-P-143 - Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of.
TT-S-735 - Standard Test Fluids, Hydrocarbon.
TT-T-548 - Toluene, Technical..
TT-X-916 - Xylene (for Use in Organic Coatings).

FSC 8010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The primer furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.4). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 The color shall be characteristic of red or brown iron oxide pigments.

3.3 Composition. The primer shall be furnished in 2 parts: Part A (Epoxy resin component) and Part B (Catalyst component). When mixed four parts by volume of Part A to one part by volume of Part B a product meeting the applicable requirements of this specification shall result.

3.3.1 Pigment. The pigment portion of the primer shall consist of the ingredients in the proportions by weight specified in Table I. When the primer is tested in accordance with section 4, the analysis shall conform to the requirements of Table II. Zinc chromate or neutral lead chromate shall not be employed alone or as a component part of any pigment.

TABLE I - Pigment composition

Characteristics	Requirements	
	Minimum	Maximum
Iron oxide	17	--
Basic lead silico chromate	51	--
Siliceous extending pigments, and anti-settling agents	--	32

3.3.2 Vehicle.

3.3.2.1 Part A (Epoxy resin component). Part A shall consist of a bisphenol type epoxy resin and pigment combined with necessary amounts of flow control agents and volatile solvents to meet the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

3.3.2.2 Part B (Catalyst component). Part B shall consist of an aliphatic polyamine-epoxy resin adduct combined with the necessary amounts of volatile solvents to meet the requirements of the specification.

3.3.2.3 Volatile content (composition L). The volatile content of both Part A and Part B when tested separately shall conform to the following requirements by volume when tested as in 4.4.5.

(a) A combination of hydrocarbons, alcohols, aldehydes, esters, ethers, and ketones having an olefinic or cyclo-olefinic type of unsaturation: 5 percent maximum.

(b) A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethyl benzene: 8 percent maximum.

(c) A combination of ethyl benzene, ketones having branched hydrocarbon structures, or toluene: 20 percent maximum.

(d) Total of a + b + c: 20 percent maximum.

3.4 Quantitative requirements.

3.4.1 Part A (Epoxy resin component). Part A shall conform to the quantitative requirements of Table II when tested as in 4.4.

TABLE II - Part A requirements

Characteristics	Minimum	Maximum
Total solids, percent by weight of Part A	62	--
Pigment, percent by weight of Part A	38	42
Vehicle solids, percent by weight of Part A	22	26
Epoxy resin, percent by weight of vehicle solids	90	--
Fineness of grind	6	--
Viscosity, Krebs Stormer, Shearing rate - 200 rpm		
Grams	125	175
Equivalent K. U.	67	77
Weight per gallon, pounds	11.2	11.5
Coarse particles and skins (retained on a 325 standard sieve), percent by weight of pigment	--	1.0
Iron oxide (Fe_2O_3), percent by weight of pigment	16	--
Basic lead silico chromate	Positive	
Zinc chromate	Negative	
Neutral lead chromate	Negative	
Total chromium as lead chromate ($PbCrO_4$), percent by weight of pigment	4.3	--
Total lead as lead sulfate ($PbSO_4$), percent by weight of pigment	32	--
Acid insoluble, percent by weight of pigment	50	53

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

3.4.2 Part B (Catalyst component). Part B shall conform to the quantitative requirements of Table III when tested as in 4.4.

TABLE III - Part B requirements

Characteristics	Minimum	Maximum
Amine nitrogen content, percent by weight of Part B	2.5	3.0
Viscosity, No. 4 Ford cup, seconds	12	20
Epoxy resin	Positive	
Weight per gallon, pounds	7.6	8.0

3.4.3 Mixed primer. The mixed primer shall conform to the quantitative requirements of Table IV when tested as in 4.4.

TABLE IV - Quantitative requirements

Characteristics	Minimum	Maximum
Viscosity, reduced, No. 4 Ford cup, seconds	15	25
Flash point, closed cup, °F.	75	--
Specular gloss, 60 degree	10	50
Drying time		
Set to touch, minutes	--	10
Dry hard, hours	--	1-1/2
Full hardness, hours	--	24

3.5 Qualitative requirements.

3.5.1 Condition in container.

3.5.1.1 Part A. When tested as in 4.4.8.1, Part A shall be free from grit, seeds, skins, abnormal thickening or livering in a freshly opened container and shall show no more pigment settling or caking than can be easily and completely reincorporated to a smooth homogeneous state.

3.5.1.2 Part B. When tested as in 4.4.8.2, Part B shall be clear and free from sediment and suspended matter when examined by transmitted light. It shall show no livering, curdling, gelling or skinning in a freshly opened full container.

3.5.2 Storage stability.

3.5.2.1 Part A. A full quart can of Part A shall show no skinning, livering, curdling, hard dry caking or tough gummy sediment when tested as in 4.4.9.1. It shall remix readily to a smooth homogeneous state, shall have a maximum viscosity of 86 K. U. and shall meet all other requirements of the specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

3.5.2.2 Part B. When tested as in 4.4.9.2, a full 8-ounce can of Part B shall be clear and free from sediment and suspended matter when examined by transmitted light. It shall show no livering, curdling, gelling, or skinning in a freshly opened container, shall have a maximum viscosity of 25 seconds and shall meet all other requirements of the specification.

3.5.3 Mixing properties. When tested as in 4.4.10, a smooth homogeneous mixture shall result. The primer shall be free from grit, seeds, skins, or lumps. After aging as in 4.4.10, the primer shall show no signs of gelation.

3.5.4 Spraying properties. When tested as in 4.4.11, the primer shall spray satisfactorily in all respects and shall show no running, sagging, or streaking. The dried film shall show no dusting, mottling, or color separation and shall present a smooth finish free from seeds.

3.5.5 Knife test. A film of primer, tested as in 4.4.12, shall adhere tightly to the test panel. It shall be difficult to furrow off with the knife and shall not flake, chip or powder. The knife cut shall show beveled edges.

3.5.6 Resistance to trichloroethylene vapors. The primer tested as in 4.4.13 shall show no blistering, flaking or peeling upon removal from the vapors. When cut as in 4.4.13, the primer shall be difficult to furrow off and shall not flake, chip, or powder. The knife cut shall show beveled edges.

3.5.7 Flexibility. A film of primer, tested as in 4.4.14, shall show no cracking or flaking.

3.5.8 Water resistance. A film of primer, tested as in 4.4.15, shall show no wrinkling or blistering immediately after removal of the panel from the water. The primer shall be no more than slightly affected when examined 2 hours after removal; and after 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness and adhesion from the portion which was not immersed. A slight whitening of the immersed portion shall be permitted.

3.5.9 Hydrocarbon fluid resistance. A film of primer, tested as in 4.4.16, shall show no blistering or wrinkling and no more than a slight whitening or softening upon removal from the fluid. After 2 hours air drying, the portion of the panel that was immersed shall be almost indistinguishable with regard to hardness, color and gloss from a panel prepared at the same time but not immersed.

3.5.10 Salt spray resistance. A film of primer, tested as in 4.4.17, and examined immediately after removal from the test, shall show no more than a trace of rusting and no more than five scattered blisters no larger than 1 mm. in diameter. The maximum amount of rusting shall be comparable to Photo 9-1 Method 6451 of Fed. Test Method Std. No. 141.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

3.5.11 Recoating. A film of primer, tested as in 4.4.18, shall show no blistering, wrinkling or other evidence of lifting. The topcoat shall be difficult to remove from the primer and the primer from the panel when cut with the knife blade.

3.5.12 Weather resistance. A film of primer, exposed as in 4.4.19, shall show no rusting, cracking, checking, flaking or loss of adhesion. The maximum amount of rusting shall be comparable to Photo 9-1 Method 6451 of Fed. Test Method Std. No. 141.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling, inspection, and testing. Unless otherwise specified, sampling, inspection and testing shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Classification of tests. Testing under this specification shall be for the purpose of:

(a) Qualification.
(b) Acceptance of individual lots.

4.3.1 Qualification testing shall consist of all tests specified in section 3 (see 6.4).

4.3.2 Acceptance testing of individual lots shall normally consist of tests for all requirements in section 3 with the exception of storage stability (see 3.5.2) and weather resistance (see 3.5.12).

4.4 Test methods.

4.4.1 Test conditions. The routine and referee testing conditions shall be in accordance with section 7, Fed. Test Method Std. No. 141, except as otherwise specified herein.

4.4.2 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141 except as otherwise specified herein. For all tests requiring the use of the mixed primer, parts A and B shall be mixed in the proportions specified in paragraph 4.4.10 and allowed to stand 30 minutes before using.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

TABLE V - Index

Item	Test Method: Applicable method in Fed. Test Method Std. No. 141	Test Method: Paragraph of this specification giving further references	Paragraph of this specification giving requirements
Pigment analysis	--	--	Tables I, II
Extraction of pigment	4021	4.4.3.2	--
Iron oxide	7141	4.4.3.2.1	Tables I, II
Basic lead silico chromate	--	4.4.3.2.2	Tables I, II
Zinc chromate	--	4.4.3.2.3	3.3.1, Table II
Neutral lead chromate	--	4.4.3.2.3	3.3.1, Table II
Total chromium	7131	4.4.3.2.4	Table II
Total lead	--	4.4.3.2.5	Table II
Acid insoluble	5271	4.4.3.2.6	Table II
Total solids	4041	--	Table II
Amine nitrogen	7391	4.4.4.1	Table III
Pigment percentage	4022	--	Table II
Isolation of vehicle	4032	--	--
Vehicle solids	4041	--	Table II
Epoxy resin	--	--	--
Part A	7401	4.4.3.1	Table II
Part B	--	4.4.4.2	Table III
Esters, ketones, alcohols and aromatic hydrocarbons	7360	4.4.5	3.3.2.3
Viscosity (Krebs-Stormer)	4281	--	Table II
Viscosity (Ford cup)	4282	4.4.6	Tables III, IV
Weight per gallon	4184	--	Tables II, III
Coarse particles & skins	4092	--	Table II
Flash point	4293	--	Table IV
Specular gloss	6101	--	Table IV
Drying time	--	4.4.7	Table IV
Set to touch	4061	--	Table IV
Dry hard	4061	--	Table IV
Full hardness	4061	--	Table IV
Condition in container	--	4.4.8	3.5.1
Part A	3011	4.4.8.1	3.5.1.1
Part B	4261	4.4.8.2	3.5.1.2
Storage stability	--	4.4.9	3.5.2
Part A	3022	4.4.9.1	3.5.2.1
Part B	4261	4.4.9.2	3.5.2.2
Mixing properties	--	4.4.10	3.5.3
Spraying properties[1]	4331	4.4.11	3.5.4
Knife test	6304	4.4.12	3.5.5
Resistance to trichloroethylene vapor	--	4.4.13	3.5.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

TABLE V - Index (continued)

Item	Test Method: Applicable method in Fed. Test Method Std. No. 141	Test Method: Paragraph of this specification giving further references	Paragraph of this specification giving requirements
Flexibility	6221	4.4.14	3.5.7
Water resistance	6011	4.4.15	3.5.8
Hydrocarbon fluid resistance	6011	4.4.16	3.5.9
Salt spray resistance	6061	4.4.17	3.5.10
Recoating	--	4.4.18	3.5.11
Weather resistance	6160	4.4.19	3.5.12

4.4.3 Analysis of Part A.

4.4.3.1 Epoxy resin. Determine the epoxy resin content on the extracted vehicle in accordance with method 7401 of Fed. Test Method Std. No. 141 for compliance with Table II.

4.4.3.2 Analysis of pigment. Extract the pigment as in method 4021 of Federal Standard 141.

4.4.3.2.1 Iron oxide (Fe_2O_3) content. Determine the iron oxide (Fe_2O_3) content on the extracted pigment in accordance with method 7141 of Fed. Test Method Std. No. 141 for compliance with Table II.

4.4.3.2.2 Basic lead silico chromate. Place a few milligrams of the isolated pigment in a smooth, highly polished agate mortar and distribute over the surface by moderate gentle grinding with a smooth agate pestle. Rub the pigment to an extremely fine powder by vigorous back and forward motion across the mortar for at least 3 minutes, and until the particles are tightly caked to the mortar. Add one drop of Nujol mineral oil and continue grinding with a rotary motion until all pigment has been suspended in oil, adding an additional drop or two if it becomes too viscous. An examination of the spectrum should show modifications of the following points of absorption which are characteristics of basic lead silico chromate pigment.

8.55 - 8.65	microns	-- shoulder
9.25	microns	-- strong broad band
11.35 - 11.8	microns	-- medium-strong broad band
12.55 and 12.87	microns	-- medium-strong bands
14.45	microns	-- weak band

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

4.4.3.2.3 Zinc chromate and neutral lead chromate. Place approximately 0.5 gram of the isolated pigment in a 50-ml. glass stoppered graduated cylinder and add 50 ml. 0.01 N NaOH. Shake and place in a water bath at 60°C. for about 30 minutes. Stopper, shake occasionally while cooling to room temperature. Filter through a double thickness of the finest filter paper, or centrifuge. Examine the color of the alkali solution. There should be no yellow color. The presence of yellow color indicates the presence of zinc chromate and/or neutral lead chromate pigments.

4.4.3.2.4 Total chromium. Determine the total chromium in the extracted pigment as lead chromate ($PbCrO_4$) in accordance with method 7131 of Fed. Test Method Std. No. 141.

4.4.3.2.5 Total lead. Place 1 gram of the pigment in a 400 ml. beaker, moisten with a few drops of alcohol, add 25 ml. of concentrated hydrochloric acid, cover and boil for 5 to 10 minutes. Dilute to 100 ml. with hot water and boil again for 5 minutes, filter the hot solution and wash with boiling water until the washings are free from lead and chlorine. Add ammonium hydroxide to the filtrate until a faint precipitate appears, then add 5 ml. of concentrated hydrochloric acid, dilute to 500 ml. and pass into the clear solution a rapid current of hydrogen sulfide until all the lead is precipitated as lead sulfide. Let the precipitate settle, filter and wash with water containing a little hydrogen sulfide. Boil the filter paper and precipitate with diluted nitric acid (1:3) until all of the lead sulfide has dissolved. Filter and wash thoroughly with hot water. To the filtrate add 10 ml. of sulfuric acid (1:1), evaporate until copious fumes of sulfur trioxide are evolved, cool, add about 75 ml. of water and then 75 ml. of 95 percent ethyl alcohol. Let stand about 1 hour, filter on a Gooch crucible, wash with diluted alcohol, dry, ignite, and weigh as lead sulfate ($PbSO_4$). Observe for compliance with Table II.

4.4.3.2.6 Acid insoluble. Determine the acid insoluble on the extracted pigment as in method 5271 of Fed. Test Method Std. No. 141.

4.4.4 Analysis of Part B.

4.4.4.1 Amine nitrogen content. Determine the amine nitrogen content as in method 7391 of Fed. Test Method Std. No. 141.

4.4.4.2 Epoxy resin. Place 4 drops of Part B in a test tube. Add about 10 drops of benzene and 10 drops of ethyl alcohol, mix and dry in an oven at 105°C. After cooling, add 1 ml. concentrated sulfuric acid and warm to about 60°C. in a water bath for 10 minutes. Cool and add 2 drops of 40 percent formaldehyde solution and allow the sample to stand a few minutes. Dilute with 10 ml. of water added all at one time. A green or blue color will form almost immediately if epoxy resins of the bisphenol type are present.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

4.4.5 Analysis of volatile (composition L). Determine solvent as in method 7360 of Fed. Test Method Std. No. 141. Samples that fail to meet the requirements of 3.3.2.3 shall be subject to further examination using a six foot silicone nitrile column (20% XF-1150 on 60 to 80 mesh Chromosorb W).

4.4.6 Viscosity (reduced). Reduce 4 parts by volume of the mixed primer with one part of xylene conforming to TT-X-916. For composition L use thinner conforming to Table VI. Test as specified in method 4282 of Fed. Test Method Std. No. 141 and check for compliance with Table IV.

TABLE VI - Thinner for Composition L

Ingredient	Percent by weight
VM&P naphtha (8% max. aromatic)	65
n-Butyl alcohol (TT-B-846)	20
Toluene (TT-T-548)	15

4.4.7 Drying time. Draw down a film of the mixed primer with a 0.0015 inch (0.0030 gap clearance) film applicator and determine drying time in accordance with method 4061 of Fed. Test Method Std. No. 141 under referee conditions and check for compliance with Table IV. The film shall be considered to have reached full hardness when it is very difficult to remove with a knife blade.

4.4.8 Condition in container.

4.4.8.1 Part A. Determine package condition of Part A on acceptance testing in accordance with method 3011 of Fed. Test Method Std. No. 141 and observe for compliance with 3.5.1.1. On qualification testing, evaluate pigment settling or caking by proceeding as in method 3011 of Fed. Test Method Std. No. 141 but do not stir. Reseal and then agitate the can for 3 minutes on a paint shaker[1/]. On re-examination of the contents, the disclosure of any gel bodies or undispersed pigment indicates unsatisfactory settling properties.

4.4.8.2 Part B. Determine package condition of Part B in accordance with method 4261 of Fed. Test Method Std. No. 141 and observe for compliance with 3.5.1.2.

4.4.9 Storage stability.

4.4.9.1 Part A. Allow a full quart can of Part A to stand undisturbed for 6 months in accordance with method 3022 of Fed. Test Method Std. No. 141 and then examine the contents. Evaluate the pigment settling as in 4.4.8.1 except agitate the can for five minutes on the paint shaker prior to re-examination. Determine viscosity and other applicable tests for compliance with 3.5.2.1.

1/ An apparatus of this type, powered by 1/4 hp motor operates at a rate of 1350 shakes per minute, and is manufactured by Red Devil Tools, Irvington, N. J.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

4.4.9.2 Part B. Allow a full eight ounce can of Part B to stand undisturbed for 6 months. At the end of this period determine the viscosity and examine the contents in accordance with method 4261 of Fed. Test Method Std. No. 141 for compliance with 3.5.2.2.

4.4.10 Mixing properties. Thoroughly mix 4 parts by volume of Part A with one part by volume of Part B and examine for compliance with 3.5.3. Place 5 ounces of the material in an eight ounce glass jar and do not agitate or disturb for 24 hours. At the end of this period examine for compliance with 3.5.3.

4.4.11 Spraying properties. Reduce the primer as in 4.4.6. Spray on a steel panel to a dry film thickness between 0.0009 and 0.0011 inch and observe for spraying properties in accordance with method 4331 of Fed. Test Method Std. No. 141 for compliance with 3.5.4. For referee test use automatic application per method 2131 of Fed. Test Method Std. No. 141.

4.4.12 Knife test. Spray the primer to a dry film thickness between 0.0009 and 0.0011 inch on a steel panel phosphoric acid etched as specified in method 2011 procedure B of Fed. Test Method Std. No. 141. Allow the panel to air dry 72 hours and then perform the knife test in accordance with method 6304 of Fed. Test Method Std. No. 141 and observe for compliance with 3.5.5.

4.4.13 Resistance to trichloroethylene vapor. Prepare a film of primer as in 4.4.12 and air dry for 72 hours. Expose the panel for one hour to trichloroethylene vapors as in paragraph 4.1.2 of method 2011 of Fed. Test Method Std. No. 141. At the end of the test period remove, and observe for compliance with 3.5.6. Allow the panel to cool to room temperature and then perform the knife test as in method 6304 of Fed. Test Method Std. No. 141 and observe for compliance with 3.5.6.

4.4.14 Flexibility. Spray a film of primer to a dry film thickness between 0.0009 and 0.0011 inch on a smooth finish steel panel prepared in accordance with method 2011 of Fed. Test Method Std. No. 141 using the aliphatic naphtha-ethylene glycol monoethyl ether mixture. The panel shall be prepared from new cold rolled carbon steel rust-free 0.010 ± 0.001 inch thick with a Rockwell 15-T maximum hardness of 82 and finished with a surface roughness of 8 to 12 micro-inches. Allow the panel to air dry 72 hours and then bend over a 1/4 inch mandrel as in method 6221 of Fed. Test Method Std. No. 141. Examine for compliance with 3.5.7.

4.4.15 Water resistance. Prepare a film of primer as in 4.4.12 and air dry 168 hours. Coat all exposed, uncoated metal surfaces with wax or other suitable coating and immerse the panel in distilled water at 23° ± 1°C. for 168 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. On removal observe the panel for compliance with 3.5.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

4.4.16 Hydrocarbon fluid resistance. Prepare a film of primer as in 4.4.12 and air dry 168 hours. Immerse in a hydrocarbon fluid conforming to TT-S-735 type III for 168 hours at 23° ± 1°C. in accordance with method 6011 of Fed. Test Method Std. No. 141. At the end of the test period remove and examine for compliance with 3.5.9.

4.4.17 Salt spray resistance. Prepare three 4 by 12 inch test panels as in 4.4.12 and air dry 72 hours. Expose the unscored panels to 5 percent salt spray for 336 hours in accordance with method 6061 of Fed. Test Method Std. No. 141. Upon removal, wash the panels gently in warm running water not more than 100°F. until free from any visible salt deposits and examine immediately for compliance with 3.5.10.

4.4.18 Recoating. Prepare 3 test panels of primer as in 4.4.12. Allow to air dry 2, 24, and 168 hours respectively and spray a coat of olive drab lustreless enamel conforming to TT-E-516 over the specimens. The enamel shall be thinned for spray as required by the specification and the dry film thickness shall be between 0.0009 and 0.0011 inch. Examine the panels for evidence of lifting after the enamel has air dried 24 hours. Allow the specimens to air dry 168 hours after recoating, cut with a knife blade and check for compliance with 3.5.11.

4.4.19 Weather resistance. Prepare two unscored 4 by 12 inch test specimens of the primer as in 4.4.12. Allow the primer to air dry for 24 hours and then apply a coat of olive drab lustreless enamel as in 4.4.18. Allow to air dry 72 hours and place on exposure for 18 months at an angle of 45° facing south in the vicinity of Washington, D. C. as in method 6160 of Fed. Test Method Std. No. 141. Examine for compliance with 3.5.12.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The primer shall be packaged, packed and marked in accordance with TT-P-143. The level of packaging shall be A or C and the level of packing shall be A, B or C as specified (see 6.2).

5.1.1 Part A (Epoxy resin component). Part A shall be packaged in one-gallon multiple friction top containers or in 5-gallon lug cover steel pails as specified. When 5-gallon steel pails are specified, they shall contain 4 gallons of epoxy resin component.

5.1.2 Part B (Catalyst component). Part B shall be packaged in 1-quart or 1-gallon multiple friction top containers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

5.2 Additional marking. In addition to markings required by TT-P-143, individual containers shall be marked Part A (Epoxy resin component) or Part B (Catalyst component) whichever is applicable. In addition individual containers shall be marked with the following information:

"CAUTION:

Avoid contact with skin and eyes.
Keep containers closed.
Avoid prolonged or repeated breathing of vapor.
Use with adequate ventilation.
Wear fresh clothing daily when contacting Part B catalyst component.

"Instructions for use. Mix Part A well; then add 1 part by volume of Part B to 4 parts by volume of Part A and mix well. Local exhaust ventilation should be provided where Part A and Part B are mixed. For spray application this material may be thinned with xylene. For composition L thin with a xylene substitute that will meet the requirements of Air Pollution Regulations. In the spray application avoid inhalation, and eye or skin contact. This material shall stand 30 minutes after mixing before using and shall be used within 8 hours after mixing."

6. NOTES

6.1 Intended use. This primer has good exterior durability, will withstand trichloroethylene degreasing vapors and is intended for use as a chemical resistant primer coat for materiel. It may be baked at 300°F. for 20 minutes or at lower temperatures for a longer period of time.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Composition required (see 1.2).
(c) Level of packaging and level of packing (see section 5).
(d) Size of container required for Part A and size container for Part B (see section 5).

6.3 The primer covered by this specification should be purchased by volume, the unit being one U. S. liquid gallon of 231 cubic inches at 68°F. (20°C.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is US Army Aberdeen Research & Development Center, Coating & Chemical Laboratory, Aberdeen Proving Ground, Maryland 21005 and information pertaining to qualification of products may be obtained from that activity.

6.5 Composition L primer should be specified for use in areas with regulations controlling the emission of solvents into the atmosphere.

Custodian:
Army - MR

Review interest:
Army - MU, MI

Preparing activity:
Army - MR

(Project No. 8010-A020)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.